

$$\text{Recall} = \frac{TP}{TP + FN}$$

recall measures how many of the actual positive cases the model

When is recall important

- Recall is important when missing a positive case is costly
- Detecting diseases
 - fraud
 - Customer churn
- } A model with high recall produces fewer false negatives

F1-score

measure the model's accuracy

Combines the precision and recall scores of a model

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 = \frac{2TP}{2TP + FP + FN}$$

calculating it using the confusion matrix

When is the F1-score important

The dataset is imbalanced

- Both false positive (FP) and false negative (FN)
- A balanced precision and recall

Interpretation: $F1 = 1$; Perfect precision and recall